

XPW Breakout v2.10 - settings by timeframe

BTCUSD, XAUUSD and SPX500 builds, TradingView Pine v6. One sheet per timeframe listing every input of the script and what to set it to. All numbers come from the calibration sweeps in CALIBRATION.md (sections 7, 13, 14 and 15) on the exports in calibration/data/: CRYPTO:BTCUSD 15m to 240m (99 days), MEXC:BTCUSDT 5m, OANDA:XAUUSD 5m to 240m, SPCFD:SPX 15m to weekly (5 months to 52 years). Costs are the VT Markets MT5 account; SPX costs are a placeholder (0.5 point spread) until the spec is confirmed.

How to apply a sheet

1. Open the chart on the symbol you trade and the sheet's timeframe. Paste the build's .pine file into the Pine editor and add it to the chart.
2. Strategy Tester > Properties > Defaults > **Reset settings**, so the header commission and slippage are the build's (BTC: cash 10.5 per contract, 500 ticks; gold: cash 0.13, 5 ticks). Bar Magnifier on if your plan has it.
3. Open the strategy inputs. Leave every row at the script default except the yellow rows of the sheet, which are the only ones that change with the timeframe.
4. Check the on-chart table: the 'Header slippage should be' row must equal the Properties slippage, and the Levels row must not read NONE.
5. Both builds share one logic; only the symbol defaults differ (costs per BTC or per ounce, sizing units, SL/TP/buffer). Do not paste a BTC sheet on a gold chart.

Summary

Symbol / TF	Verdict	Rows that change	Trades	Net	PF	Max DD
BTCUSD 5m and below	do not trade	-	109	-6,519	0.49	-
BTCUSD 15m	trade	Geometry Pct, SL 1.0%, TP 2.5%, Script trail	29	+5,957	2.30	2,387
BTCUSD 30m	trade (Donchian 50 variant better)	Script trail or Donchian 50	24 24	+5,660 +7,585	1.91 3.90	4,625 1,277
BTCUSD 60m	shipped defaults	none	30	+10,004	2.35	2,956
BTCUSD 240m	low confidence	Donchian 20 instead of pivots	27	+5,850	1.50	5,683
XAUUSD 5m	do not trade	-	48	-59/oz	-	-
XAUUSD 10m	trade	Donchian 20 instead of pivots	124	+141/oz	1.27	85
XAUUSD 15m	shipped defaults	none	101	+213/oz	1.49	94
XAUUSD 30m	shipped defaults	none	54	+194/oz	1.65	104
XAUUSD 60m	trade	Donchian 20 instead of pivots	49	+238/oz	1.52	100
XAUUSD 240m	trade	Donchian 5 instead of pivots	64	+942/oz	1.73	236
SPX500 5m and below	not tested	-	-	-	-	-
SPX500 15m	borderline	Pct 0.1% / 0.25%, pivot bars 5, buffer 0.5	173	+400/pt	1.49	102
SPX500 30m	trade	Geometry ATR (3.0 / 3R), pivot bars 8, buffer 2.0	78	+764/pt	1.67	296
SPX500 60m	shipped defaults	none	143	+1,320/pt	1.59	333
SPX500 120m	shipped defaults	none	151	+1,329/pt	1.57	225
SPX500 180m	trade	Geometry ATR (3.0 / 3R), pivot bars 8	82	+1,838/pt	2.03	416
SPX500 240m	trade, few trades	Geometry ATR (3.0 / 3R), pivot bars 8, buffer 1.0, trail off	34	+3,450/pt	2.42	803
SPX500 daily	trade	Geometry ATR 2.0 / 2.5R, pivot bars 8, buffer 0.5, trail off	56	+4,767/pt	2.66	451
SPX500 weekly	context only	daily block	66	+2,093/pt	1.53	1,027

Net per 1 BTC, per 1 oz (x100 for one gold lot) or per 1 SPX contract at \$1 per index point, fixed size, after costs. Trade counts are over the whole export: 99 days of BTC, 2 to 12 weeks of gold, 5 months (15m) to 52 years (weekly) of SPX.

Reading the evidence

Shipped defaults (BTC 60m; gold 15m and 30m; SPX 60m and 120m) are the settings chosen to be positive across timeframes rather than best on one. SPX has no single geometry that works from 30m to daily: intraday wants a percent stop, 180m and above an ATR stop, so the sheets switch Geometry per timeframe. They are the only rows validated by a first-half / second-half split and, for gold, a walk-forward (5 of 5 timeframes positive out of sample).

Donchian rows replace the pivot with a rolling prior-N high/low, so the buy stop and sell stop move every bar. They are recommended only where both halves of the export were positive and the level source was robust across the geometry grid. The wrong length is destructive (Donchian 50 on BTC 240m loses in every band).

Follow versus Recheck (what a resting stop does when its level is replaced) is a wash across the grid: median PF difference under 0.1, more trades with Follow. Follow is the default because every calibration number was measured with it.

Single-cell alternatives in the notes are the best plateau cell of that timeframe. They have 18 to 39 trades and the gold walk-forward showed per-timeframe picks failing out of sample 2 to 3 times in 5; use them with that in mind.

Not modelled: swaps and financing, the gold maintenance break and weekend gap, exchange funding, and live spread widening at news.

BTCUSD - 5m and below: DO NOT TRADE

Script: pine/XPW_Breakout_v2.10_BTCUSD.pine. Net figures per 1 BTC, VT Markets costs (\$31 round trip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: Defaults on the MEXC 5m export: 109 trades, -6,519, PF 0.49, in a week where 15m/30m/60m were positive.

- No settings sheet. If a 5m chart must be used, the only band with evidence (one week) is Geometry Pct, SL 1.0%, TP 1.0% to 3.0%, pivot bars 8. Treat it as untested.

BTCUSD - 15m: Trade with the 15m block

Script: pine/XPW_Breakout_v2.10_BTCUSD.pine. Net figures per 1 BTC, VT Markets costs (\$31 round trip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: 29 trades, +5,957, PF 2.30, max DD 2,387 (sweep, CFD costs within \$4 of VT). Every grid neighbour positive.

- Percent geometry wins on 15m: SL 1.0% is about 4 ATR there; the ATR 3.0 band is positive but thinner (36 trades, +4,509).

- Alternative, same script defaults with a rolling level: Pivot levels OFF, Donchian ON, length 20, Geometry ATR 3.0 / 3R, buffer 0.5, Emulator trail, Follow: 37 trades, +5,877, PF 2.56, max DD 1,141; first half +1,801 / second half +3,661. The pivot at the ATR defaults loses the first half (-594 / +4,908).

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 10.5 USD	Header default. Properties > Defaults > Reset settings restores it.
Strategy Tester Properties	Properties > Slippage	500 ticks	= 5 USD at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	10.5	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	5.0	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	0.1 BTC	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	2.0 BTC	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	0.01	
4. Geometry	Geometry	Pct (default ATR)	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	3.0	
4. Geometry	TP, R multiple of SL (ATR mode)	3.0	
4. Geometry	TP as percent of entry (Pct mode)	2.5 (default 0.25)	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	1.0 (default 0.1)	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	3	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	0.5	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	Max trades per day (0 = off)	0	
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	
			Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Script (default Emulator)	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

BTCUSD - 30m: Trade, prefer the Donchian 50 variant

Script: pine/XPW_Breakout_v2.10_BTCUSD.pine. Net figures per 1 BTC, VT Markets costs (\$31 round trip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: Pivot, ATR 3.0 / 3R, BarsN 3, buffer 0.5, Script trail: 24 trades, +5,660, PF 1.91, max DD 4,625.

- Do not use pivot bars 5 on 30m (30 trades, +1,411 CFD / -486 exchange). 3 or 8 only.

- Better variant: Pivot levels OFF, Donchian ON, length 50, Geometry ATR 3.0 / 3R, buffer 0.5, Emulator trail: 24 trades, +7,585, PF 3.90, max DD 1,277; first half +651 / second half +3,979, but only 11 trades per half. Follow and Recheck give the same result here.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 10.5 USD	Header default. Properties > Defaults > Reset settings restores it.
Strategy Tester Properties	Properties > Slippage	500 ticks	= 5 USD at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	10.5	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	5.0	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	0.1 BTC	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	2.0 BTC	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	0.01	
4. Geometry	Geometry	ATR	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	3.0	
4. Geometry	TP, R multiple of SL (ATR mode)	3.0	
4. Geometry	TP as percent of entry (Pct mode)	0.25	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.1	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	3	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	0.5	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	
7. Filters	Max consecutive losses per level (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Script (default Emulator)	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

BTCUSD - 60m: SHIPPED DEFAULTS - change nothing

Script: pine/XPW_Breakout_v2.10_BTCUSD.pine. Net figures per 1 BTC, VT Markets costs (\$31 round trip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: 30 trades, +10,004, PF 2.35, max DD 2,956; first half +397 / second half +8,883. Recheck: 31 trades, +9,937, PF 2.29.

- Sensitive to pivot bars: 5 and 8 drop to about +2,400. Keep 3.

- Donchian does not help here at the defaults (Donchian 10: +7,387 with a negative first half; Donchian 20: +4,023). The pivot is the only level source positive in both halves on 60m.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 10.5 USD	Header default. Properties > Defaults > Reset settings restores it.
Strategy Tester Properties	Properties > Slippage	500 ticks	= 5 USD at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	10.5	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	5.0	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	0.1 BTC	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	2.0 BTC	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	0.01	
4. Geometry	Geometry	ATR	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	3.0	
4. Geometry	TP, R multiple of SL (ATR mode)	3.0	
4. Geometry	TP as percent of entry (Pct mode)	0.25	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.1	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	3	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	0.5	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

BTCUSD - 240m: Low confidence - context timeframe

Script: pine/XPW_Breakout_v2.10_BTCUSD.pine. Net figures per 1 BTC, VT Markets costs (\$31 round trip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: Donchian 20, ATR 3.0 / 3R, buffer 0.5, Emulator trail, Follow: 27 trades, +5,850, PF 1.50, max DD 5,683; first half +885 / second half +5,591.

- 99 days of 240m bars give 20 to 46 trades per cell and the sign flips between neighbours; the pivot defaults are +1,528 (Recheck) or -2,708 (Follow). Donchian 20 is the only level source positive in both halves. Every other Donchian length loses on 240m (length 5: -13,624).

- Use this sheet only with a larger position cap than you would regret; more history is needed before 240m counts as calibrated.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 10.5 USD	Header default. Properties > Defaults > Reset settings restores it.
Strategy Tester Properties	Properties > Slippage	500 ticks	= 5 USD at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	10.5	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	5.0	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	0.1 BTC	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	2.0 BTC	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	0.01	
4. Geometry	Geometry	ATR	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	3.0	
4. Geometry	TP, R multiple of SL (ATR mode)	3.0	
4. Geometry	TP as percent of entry (Pct mode)	0.25	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.1	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	off (default on)	
5. Levels	pivot bars each side	3	BarsN.
5. Levels	Donchian levels (prior N bars)	on (default off)	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	0.5	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

XAUUSD - 5m: DO NOT TRADE

Script: pine/XPW_Breakout_v2.10_XAUUSD.pine. Net figures per 1 oz, VT Markets raw costs; multiply by 100 for one MT5 lot. Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: Defaults: 48 trades, -59 per oz. Every geometry that survives costs elsewhere is negative on 5m.

- No settings sheet.

XAUUSD - 10m: Trade with Donchian 20

Script: pine/XPW_Breakout_v2.10_XAUUSD.pine. Net figures per 1 oz, VT Markets raw costs; multiply by 100 for one MT5 lot. Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: Donchian 20, ATR 1.5 / 2.5R, buffer 1.0, Emulator trail, Follow: 124 trades, +141 per oz, PF 1.27, max DD 85; first half +109 / second half +37. Most robust level source on 10m (94% of the grid positive).

- Pivot defaults on 10m: 134 trades, +99, PF 1.16, second half +8 (a coin flip). Donchian 10 with Recheck is the higher net (+182, PF 1.36, 111 trades, +148 / +33).

- Single-cell alternative on pivots: Geometry Pct, SL 0.5%, TP 1.25%, pivot bars 8, buffer 0.5, trail OFF: 23 trades, +452, PF 3.06, max DD 44.

Thin: the walk-forward's first-half pick on 10m made +80 out of sample.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.13 USD	Header default. Properties > Defaults > Reset settings restores it.
Strategy Tester Properties	Properties > Slippage	5 ticks	= 0.05 USD/oz at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.13	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.05	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	10 oz	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	50 oz	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	ATR	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	1.5	
4. Geometry	TP, R multiple of SL (ATR mode)	2.5	
4. Geometry	TP as percent of entry (Pct mode)	0.25	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.1	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	off (default on)	
5. Levels	pivot bars each side	3	BarsN.
5. Levels	Donchian levels (prior N bars)	on (default off)	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	1.0	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	Max trades per day (0 = off)	0	
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	
			Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

XAUUSD - 15m: SHIPPED DEFAULTS - change nothing

Script: pine/XPW_Breakout_v2.10_XAUUSD.pine. Net figures per 1 oz, VT Markets raw costs; multiply by 100 for one MT5 lot. Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: 101 trades, +213 per oz, PF 1.49, max DD 94; first half +31 / second half +177; walk-forward out of sample +177. Recheck: 81 trades, +216, PF 1.65.

- The pivot is the best and most robust level source on 15m (90 to 96% of the grid positive); every Donchian length is behind it.
 - Single-cell alternative: ATR 3.0 / 2.5R, pivot bars 3, buffer 2.0, trail OFF: 18 trades, +551, PF 3.46, max DD 56. Fewer trades, and the 15m first-half plateau pick lost 42 out of sample, so prefer the defaults.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.13 USD	Header default. Properties > Defaults > Reset settings restores it.
Strategy Tester Properties	Properties > Slippage	5 ticks	= 0.05 USD/oz at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.13	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.05	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	10 oz	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	50 oz	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	ATR	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	1.5	
4. Geometry	TP, R multiple of SL (ATR mode)	2.5	
4. Geometry	TP as percent of entry (Pct mode)	0.25	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.1	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	3	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	1.0	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

XAUUSD - 30m: SHIPPED DEFAULTS - change nothing

Script: pine/XPW_Breakout_v2.10_XAUUSD.pine. Net figures per 1 oz, VT Markets raw costs; multiply by 100 for one MT5 lot. Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: 54 trades, +194 per oz, PF 1.65, max DD 104; first half +71 / second half +79; walk-forward out of sample +79. Recheck: 44 trades, +92, PF 1.34 (-3 / +51), so keep Follow here.

- Pivot is the most robust source on 30m (82 to 84% of the grid positive); Donchian 20 is close (+109, PF 1.28) and the others are behind.

- Single-cell alternative: ATR 3.0 / 1.5R, pivot bars 3, buffer 2.0, Emulator trail: 35 trades, +389, PF 6.20, max DD 40. The 30m first-half plateau pick lost 13 out of sample; treat as optional.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.13 USD	Header default. Properties > Defaults > Reset settings restores it.
Strategy Tester Properties	Properties > Slippage	5 ticks	= 0.05 USD/oz at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.13	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.05	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	10 oz	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	50 oz	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	ATR	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	1.5	
4. Geometry	TP, R multiple of SL (ATR mode)	2.5	
4. Geometry	TP as percent of entry (Pct mode)	0.25	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.1	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	3	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	1.0	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	
			Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

XAUUSD - 60m: Trade with Donchian 20

Script: pine/XPW_Breakout_v2.10_XAUUSD.pine. Net figures per 1 oz, VT Markets raw costs; multiply by 100 for one MT5 lot. Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: Donchian 20, ATR 1.5 / 2.5R, buffer 1.0, Emulator trail, Follow: 49 trades, +238 per oz, PF 1.52, max DD 100; first half +155 / second half +93.

- Pivot defaults on 60m: 55 trades, +171, PF 1.31, max DD 146, but the first half is -5 (walk-forward out of sample +186). Donchian 20 is positive in both halves with a smaller drawdown.

- Donchian 50 is the highest PF on 60m: 33 trades, +258, PF 2.14, max DD 93 (+174 / +30), and 84% of the grid positive with Recheck. Fewer trades; either length is defensible.

- Single-cell alternative on pivots: ATR 3.0 / 2.5R, pivot bars 3, buffer 0.5, Script trail: 39 trades, +579, PF 2.50, max DD 106. The 60m first-half best cell lost 396 out of sample: single cells on 60m gold are not trustworthy.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.13 USD	Header default. Properties > Defaults > Reset settings restores it.
Strategy Tester Properties	Properties > Slippage	5 ticks	= 0.05 USD/oz at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.13	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.05	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	10 oz	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	50 oz	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	ATR	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	1.5	
4. Geometry	TP, R multiple of SL (ATR mode)	2.5	
4. Geometry	TP as percent of entry (Pct mode)	0.25	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.1	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	off (default on)	
5. Levels	pivot bars each side	3	BarsN.
5. Levels	Donchian levels (prior N bars)	on (default off)	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	1.0	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.

Group	Input (as shown in the script)	Set to	Note
7. Filters	Cooltdown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

XAUUSD - 240m: Trade with Donchian 5

Script: pine/XPW_Breakout_v2.10_XAUUSD.pine. Net figures per 1 oz, VT Markets raw costs; multiply by 100 for one MT5 lot. Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

- Expected on the calibration data:** Donchian 5, ATR 1.5 / 2.5R, buffer 1.0, Emulator trail, Follow: 64 trades, +942 per oz, PF 1.73, max DD 236; first half +714 / second half +123. Donchian 5 and 10 are 92 to 99% positive across the grid on 240m.
- Pivot defaults on 240m: 53 trades, +487, PF 1.40, max DD 422 (Recheck: 47 trades, +648, PF 1.59, +283 / +365).
 - Pivot alternative with a tighter target: ATR 3.0 / 1.5R, pivot bars 3, buffer 1.0, Script trail: 34 trades, +1,526, PF 2.69, max DD 176; walk-forward out of sample +435.
 - Swaps are not modelled: a 240m hold often spans the rollover, and gold swaps (triple on Wednesday) can exceed the round-trip cost.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.13 USD	Header default. Properties > Defaults > Reset settings restores it.
Strategy Tester Properties	Properties > Slippage	5 ticks	= 0.05 USD/oz at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.13	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.05	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	10 oz	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	50 oz	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	ATR	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	1.5	
4. Geometry	TP, R multiple of SL (ATR mode)	2.5	
4. Geometry	TP as percent of entry (Pct mode)	0.25	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.1	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	off (default on)	
5. Levels	pivot bars each side	3	BarsN.
5. Levels	Donchian levels (prior N bars)	on (default off)	A rolling level that moves every bar.
5. Levels	Donchian length	5 (default 20)	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	1.0	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	Max trades per day (0 = off)	0	
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	
			Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

SPX500 (SPCFD:SPX) - 5m and below: NOT TESTED - do not trade

Script: pine/XPW_Breakout_v2.10_SPX500.pine. Net figures per 1 contract at \$1 per index point, cash session, placeholder costs (0.5 pt spread, 0.1 pt slip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: The 1m export covers 2.5 months and the 15s / 1s exports one day: not enough to calibrate, and every lower timeframe tested on BTC and gold lost.

- No settings sheet.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.25 (index points)	Half of a 0.5 point spread; placeholder until the VT Markets SPX500 spec is confirmed.
Strategy Tester Properties	Properties > Slippage	10 ticks	= 0.10 index points at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.25 pt	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.1 pt	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	1 contract	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	10 contracts	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	Pct	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	3.0	
4. Geometry	TP, R multiple of SL (ATR mode)	3.0	
4. Geometry	TP as percent of entry (Pct mode)	1.0	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.5	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	3	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	2.0	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	

Group	Input (as shown in the script)	Set to	Note
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

SPX500 (SPCFD:SPX) - 15m: Borderline - verify with Bar Magnifier first

Script: pine/XPW_Breakout_v2.10_SPX500.pine. Net figures per 1 contract at \$1 per index point, cash session, placeholder costs (0.5 pt spread, 0.1 pt slip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: Pct SL 0.1% / TP 0.25%, pivot bars 5, buffer 0.5, Emulator trail, Follow: 173 trades, +400, PF 1.49, max DD 102; first half +282 / second half +118. Donchian 10 instead of pivots: 272 trades, +717, PF 1.55 (+239 / +476).
 - One trade in four resolves inside the fill bar (the stop is 0.8 ATR on 15m), so part of this result is the emulator's path assumption. Run it in the Strategy Tester with Bar Magnifier ON before trusting it; if the PF holds there, it is real.
 - Every wider geometry was negative in the second half of the 15m export (5 months). 15m SPX is the weakest timeframe here.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.25 (index points)	Half of a 0.5 point spread; placeholder until the VT Markets SPX500 spec is confirmed.
Strategy Tester Properties	Properties > Slippage	10 ticks	= 0.10 index points at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.25 pt	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.1 pt	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	1 contract	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	10 contracts	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	Pct	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	3.0	
4. Geometry	TP, R multiple of SL (ATR mode)	3.0	
4. Geometry	TP as percent of entry (Pct mode)	0.25 (default 1.0)	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.1 (default 0.5)	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	5 (default 3)	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	0.5 (default 2.0)	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.

Group	Input (as shown in the script)	Set to	Note
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

SPX500 (SPCFD:SPX) - 30m: Trade with the ATR block

Script: pine/XPW_Breakout_v2.10_SPX500.pine. Net figures per 1 contract at \$1 per index point, cash session, placeholder costs (0.5 pt spread, 0.1 pt slip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: ATR 3.0 / 3R, pivot bars 8, buffer 2.0, Emulator trail, Follow: 78 trades, +764, PF 1.67, max DD 296; first half +573 / second half +171; 1% fill-bar exits. 9 months of data.

- The SL/TP ATR inputs stay at their defaults (3.0 / 3R); only Geometry switches to ATR and pivot bars to 8.

- Alternative with more trades: Pct SL 0.25% / TP 0.5%, Donchian 10 instead of pivots, buffer 0.5, Recheck: 237 trades, +809, PF 1.34, max DD 385 (+563 / +227), 15% fill-bar exits.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.25 (index points)	Half of a 0.5 point spread; placeholder until the VT Markets SPX500 spec is confirmed.
Strategy Tester Properties	Properties > Slippage	10 ticks	= 0.10 index points at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.25 pt	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.1 pt	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	1 contract	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	10 contracts	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	ATR (default Pct)	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	3.0	
4. Geometry	TP, R multiple of SL (ATR mode)	3.0	
4. Geometry	TP as percent of entry (Pct mode)	1.0	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.5	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	8 (default 3)	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	2.0	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

SPX500 (SPCFD:SPX) - 60m: SHIPPED DEFAULTS - change nothing

Script: pine/XPW_Breakout_v2.10_SPX500.pine. Net figures per 1 contract at \$1 per index point, cash session, placeholder costs (0.5 pt spread, 0.1 pt slip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: 143 trades, +1,320, PF 1.59, max DD 333; first half +917 / second half +437; 10% fill-bar exits. 18 months of data.

- Recheck instead of Follow halves the trade count (80 trades, +750, PF 1.55) and loses the second half (+10): keep Follow on SPX.

- Alternative: Pct SL 0.25% / TP 0.5%, pivot bars 5, buffer 0.5, Recheck: 194 trades, +1,065, PF 1.54, max DD 179 (+733 / +298), but 26% of exits are inside the fill bar.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.25 (index points)	Half of a 0.5 point spread; placeholder until the VT Markets SPX500 spec is confirmed.
Strategy Tester Properties	Properties > Slippage	10 ticks	= 0.10 index points at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.25 pt	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.1 pt	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	1 contract	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	10 contracts	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	Pct	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	3.0	
4. Geometry	TP, R multiple of SL (ATR mode)	3.0	
4. Geometry	TP as percent of entry (Pct mode)	1.0	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.5	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	3	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	2.0	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

SPX500 (SPCFD:SPX) - 120m: SHIPPED DEFAULTS - change nothing

Script: pine/XPW_Breakout_v2.10_SPX500.pine. Net figures per 1 contract at \$1 per index point, cash session, placeholder costs (0.5 pt spread, 0.1 pt slip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: 151 trades, +1,329, PF 1.57, max DD 225; first half +594 / second half +676; 15% fill-bar exits. 2.7 years of data.

- Alternative: Geometry ATR, SL 1.5 ATR, TP 3R, pivot bars 3, buffer 1.0, Emulator trail: 183 trades, +1,696, PF 1.48, max DD 455 (+1,218 / +460), 6% fill-bar exits.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.25 (index points)	Half of a 0.5 point spread; placeholder until the VT Markets SPX500 spec is confirmed.
Strategy Tester Properties	Properties > Slippage	10 ticks	= 0.10 index points at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.25 pt	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.1 pt	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	1 contract	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	10 contracts	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	Pct	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	3.0	
4. Geometry	TP, R multiple of SL (ATR mode)	3.0	
4. Geometry	TP as percent of entry (Pct mode)	1.0	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.5	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	3	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	2.0	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	
7. Filters	Max consecutive losses per level (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

SPX500 (SPCFD:SPX) - 180m: Trade with the ATR block

Script: pine/XPW_Breakout_v2.10_SPX500.pine. Net figures per 1 contract at \$1 per index point, cash session, placeholder costs (0.5 pt spread, 0.1 pt slip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: ATR 3.0 / 3R, pivot bars 8, buffer 2.0, Emulator trail, Follow: 82 trades, +1,838, PF 2.03, max DD 416; first half +1,239 / second half +577; no fill-bar exits. 3.6 years of data.

- Recheck: 70 trades, +1,898, PF 2.22, max DD 457 (+1,294 / +581). Either policy.

- The shipped percent block fails here (-30 over the file); use the ATR block.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.25 (index points)	Half of a 0.5 point spread; placeholder until the VT Markets SPX500 spec is confirmed.
Strategy Tester Properties	Properties > Slippage	10 ticks	= 0.10 index points at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.25 pt	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.1 pt	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	1 contract	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	10 contracts	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	ATR (default Pct)	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	3.0	
4. Geometry	TP, R multiple of SL (ATR mode)	3.0	
4. Geometry	TP as percent of entry (Pct mode)	1.0	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.5	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	8 (default 3)	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	2.0	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	
7. Filters	Max consecutive losses per level (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	on	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

SPX500 (SPCFD:SPX) - 240m: Trade with the ATR block, few trades

Script: pine/XPW_Breakout_v2.10_SPX500.pine. Net figures per 1 contract at \$1 per index point, cash session, placeholder costs (0.5 pt spread, 0.1 pt slip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: ATR 3.0 / 3R, pivot bars 8, buffer 1.0, trail OFF, Follow: 34 trades, +3,450, PF 2.42, max DD 803; first half +1,719 / second half +1,538. 5.4 years of data, so about 6 trades a year.

- Alternative with more trades: Geometry ATR, SL 1.5 ATR, TP 3R, pivot bars 3, buffer 1.0, Emulator trail: 189 trades, +2,028, PF 1.40, max DD 1,031 (+1,369 / +612).

- Every percent stop below 0.5% is a fill-bar artefact on 240m and above (46 to 96% same-bar exits) and is excluded.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.25 (index points)	Half of a 0.5 point spread; placeholder until the VT Markets SPX500 spec is confirmed.
Strategy Tester Properties	Properties > Slippage	10 ticks	= 0.10 index points at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.25 pt	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.1 pt	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	1 contract	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	10 contracts	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	ATR (default Pct)	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	3.0	
4. Geometry	TP, R multiple of SL (ATR mode)	3.0	
4. Geometry	TP as percent of entry (Pct mode)	1.0	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.5	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	8 (default 3)	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	1.0 (default 2.0)	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	off (default on)	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

SPX500 (SPCFD:SPX) - Daily: Trade with the daily ATR block

Script: pine/XPW_Breakout_v2.10_SPX500.pine. Net figures per 1 contract at \$1 per index point, cash session, placeholder costs (0.5 pt spread, 0.1 pt slip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: ATR 2.0 / 2.5R, pivot bars 8, buffer 0.5, trail OFF, Follow: 56 trades, +4,767, PF 2.66, max DD 451; first half +1,602 / second half +3,090; no fill-bar exits. 10.8 years of data, about 5 trades a year.

- Alternative: ATR 3.0 / 3R, pivot bars 8, buffer 2.0, Emulator trail: 55 trades, +3,039, PF 4.29, max DD 296 (+938 / +2,024).

- Positions hold for days: overnight financing on an index CFD is not modelled and will matter at this timeframe.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.25 (index points)	Half of a 0.5 point spread; placeholder until the VT Markets SPX500 spec is confirmed.
Strategy Tester Properties	Properties > Slippage	10 ticks	= 0.10 index points at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.25 pt	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.1 pt	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	1 contract	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	10 contracts	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	ATR (default Pct)	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	2.0 (default 3.0)	
4. Geometry	TP, R multiple of SL (ATR mode)	2.5 (default 3.0)	
4. Geometry	TP as percent of entry (Pct mode)	1.0	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.5	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	8 (default 3)	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	0.5 (default 2.0)	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	
7. Filters	Max consecutive losses per level (0 = off)	0	

Group	Input (as shown in the script)	Set to	Note
7. Filters	same-level tolerance, ATR mult	0.25	
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	off (default on)	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	

SPX500 (SPCFD:SPX) - Weekly: Context only

Script: pine/XPW_Breakout_v2.10_SPX500.pine. Net figures per 1 contract at \$1 per index point, cash session, placeholder costs (0.5 pt spread, 0.1 pt slip). Rows shaded yellow differ from the script defaults; everything else is left as the script loads it.

Expected on the calibration data: Same block as daily: 66 trades, +2,093, PF 1.53, max DD 1,027 over 52 years (about one trade a year); first half +359 / second half +1,839.

- Not a trading timeframe for this system. Monthly and quarterly exports were stored but not swept: a 3-bar pivot needs 7 months to confirm.

Group	Input (as shown in the script)	Set to	Note
Strategy Tester Properties	Properties > Commission	cash per contract, 0.25 (index points)	Half of a 0.5 point spread; placeholder until the VT Markets SPX500 spec is confirmed.
Strategy Tester Properties	Properties > Slippage	10 ticks	= 0.10 index points at mintick 0.01.
1. Execution	Execution	Tick	BarClose = v2.01 bar-close behaviour. Historical bars are identical either way.
1. Execution	Trading System (1 = enabled)	1	0 disables new entries.
1. Execution	Direction	Both	
1. Execution	Arm latch (buffer arms only, stop then rests)	on	Keep on. Off = v2.01 arming, which cancels the stop on every tick inside the buffer.
1. Execution	Latched stop when the level moves	Follow	Follow = the resting stop is re-priced to a new level at once (calibrated policy). Recheck = new level must pass the buffer again.
1. Execution	Max level distance, ATR mult (0 = off)	0	Must be larger than the buffer if used.
2. Costs (model, mirrors the header)	Commission % per side	0	Exchange feeds: 0.05. Then set the header commission type to percent.
2. Costs (model, mirrors the header)	Commission USD per contract per side	0.25 pt	VT Markets: BTC half of the \$21 spread; gold half of the \$0.25 spread + \$0.006/oz commission.
2. Costs (model, mirrors the header)	Slippage USD per side	0.1 pt	Header slippage (ticks) must equal this / mintick: 500 for BTC, 5 for gold. The table prints it.
2. Costs (model, mirrors the header)	Cost gate (skip when RT cost > x% of net TP)	on	
2. Costs (model, mirrors the header)	max round-trip cost, % of net TP	20	
2. Costs (model, mirrors the header)	Widen TP by both commissions (net target = input)	on	
2. Costs (model, mirrors the header)	Reserve round-trip cost in risk sizing	on	
3. Sizing	Risk percent of equity per trade	1.0	Risk sizing was spot-checked, not swept; the sweep used 1 BTC / 1 oz fixed.
3. Sizing	Risk-based sizing (off = FixedQty)	on	
3. Sizing	FixedQty fallback	1 contract	Gold: 100 oz = 1 MT5 lot.
3. Sizing	HARD qty cap	10 contracts	The cap binding means the risk inputs are wrong for the account, not a feature.
3. Sizing	Qty step (rounds down)	1	
4. Geometry	Geometry	ATR (default Pct)	Pct = percent-of-entry SL and TP (v2.01 style).
4. Geometry	SL, ATR mult (ATR mode)	2.0 (default 3.0)	
4. Geometry	TP, R multiple of SL (ATR mode)	2.5 (default 3.0)	
4. Geometry	TP as percent of entry (Pct mode)	1.0	Only read in Pct mode.
4. Geometry	SL as percent of entry (Pct mode)	0.5	Only read in Pct mode. 0.1% on BTC sits inside one bar: do not use it there.
4. Geometry	ATR length	14	
5. Levels	Pivot levels (v2.01)	on	
5. Levels	pivot bars each side	8 (default 3)	BarsN.
5. Levels	Donchian levels (prior N bars)	off	A rolling level that moves every bar.
5. Levels	Donchian length	20	Only read when Donchian is on.
5. Levels	Previous-day high/low	off	
5. Levels	Entry buffer anchor	ATR	
5. Levels	Buffer, ATR mult	0.5 (default 2.0)	Price must be this far from the level to ARM the stop.
5. Levels	Buffer, percent of the Pct-mode TP (parity mode)	50	Only for the v2.01 acceptance test.
6. Entry trigger	Trigger	Stop	Stop = pending stop straddle (the calibrated one). CloseConfirm / Retest were not swept.
6. Entry trigger	CloseConfirm: max close beyond level, ATR mult	0.5	
6. Entry trigger	Retest: bars the limit stays valid	5	
6. Entry trigger	Retest: limit offset inside the level, ATR mult	0	
7. Filters	Chart EMA trend filter, length (0 = off)	0	Not swept; the calibration ran without filters.
7. Filters	Cooldown bars after an exit (0 = off)	0	
7. Filters	Max trades per day (0 = off)	0	
7. Filters	Max consecutive losses per level (0 = off)	0	
7. Filters	same-level tolerance, ATR mult	0.25	

Group	Input (as shown in the script)	Set to	Note
7. Filters	Compression gate (arm only when coiled)	off	
7. Filters	compression lookback	200	
7. Filters	arm below width percentile	30	
8. Time	Start hour (0 = off)	0	Chart / broker server time.
8. Time	End hour (0 = off)	0	
8. Time	Trade days (1=Sun .. 7=Sat)	1234567	Gold: the daily maintenance break and weekend gap are not modelled.
9. Exits	Ratchet trail	off (default on)	
9. Exits	Trail execution	Emulator	Emulator = broker-emulator trail (intrabar on history, per tick live) = engine 'tick'. Script = v2.01 close-anchored ratchet = engine 'bar'.
9. Exits	Trail anchor	ATR	
9. Exits	Trail trigger, ATR mult	1.0	
9. Exits	Trail distance, ATR mult	1.5	
9. Exits	Trail trigger, percent of TP (parity mode)	7	v2.01 parity only.
9. Exits	Trail distance, percent of TP (parity mode)	5	v2.01 parity only.
9. Exits	Breakeven step (stop to fill +/- costs)	off	Not swept.
9. Exits	BE trigger, R multiple of SL	1.0	
9. Exits	Time stop, bars (0 = off)	0	Not swept.
10. Display	Cost / state table	on	Check the 'Header slippage should be' row after any cost change.
10. Display	Plot levels, TP and trail	on	
10. Display	Plot trend EMA	off	
10. Display	Mark realtime bars (Tick mode)	on	